

TEAM ID	SWTID-2026-8806
PROJECT TITLE	COSMETIC INSIGHTS: NAVIGATING COSMETICS TRENDS AND CONSUMER INSIGHTS WITH TABLEAU
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Here is the rewritten text:

Step 1: Load the Cosmetics Dataset into Tableau

First we need to open Tableau Desktop.

Then we click on Connect. Select Microsoft Excel, which is the file format of our cosmetics dataset.

We choose the cosmetics dataset. Make sure all the columns are imported correctly.

These columns are:

- * Product Name
- * Brand
- * Category
- * Price
- * Rank
- * Combination
- * Dry
- * Normal
- * Oily
- * Sensitive

We also need to check the data types to make sure they are correct.

The Product Name, Brand and Category should be strings.

The Price and Rank should be numbers.

The Skin Type fields should be either numbers or booleans.

Step 2: Clean the Data

Now we need to remove any duplicate records if they exist.

We also need to check for any values and rename some fields to make them easier to read.

We should verify the fields like Price and Rank to make sure they are correct.

Finally we save the cleaned data source.

Step 3: Create Filters for the Dashboard

We need to create a Category Filter, which's a Label Filter.

To do this we drag the Category to the Filters shelf. Select all the categories.

Then we right-click on the Category filter. Choose Show Filter.

We can change the filter type to either a Single Value Dropdown or a Multiple Value Dropdown.

The cosmetics dataset and the Category Filter allow users to dynamically select categories for focused analysis.

Step 4: Create a Field for All Skin Types Count

To do this we go to Analysis and create a new calculated field.

We name this field ALL SKIN TYPES COUNT.

Then we enter the following formula:

$$\text{SUM}([\text{Combination}]).\text{Sum}([\text{dry}]).\text{Sum}([\text{normal}]) + \text{SUM}([\text{Sensitive}])$$

This calculated field, the ALL SKIN TYPES COUNT calculates the products suitable for multiple skin types.

It helps us analyze the product availability of the cosmetics dataset.

Step 5: Analyze Product Distribution Across Categories

We create a visualization by opening a new worksheet.

Then we drag the Category to the Rows and the Product Name to the Columns.

We change the aggregation to COUNT(Product Name). Select a Bar Chart.

The result of this analysis displays the product distribution across categories of the cosmetics dataset.

Step 6: Calculate Category-wise Total Revenue

If the revenue is not available we can use the following formula:

SUM([Price])

To do this we drag the Category to the Rows and the Price to the Columns.

We set the aggregation to SUM. Sort it in descending order.

Then we add labels to the visualization.

The result of this analysis shows the revenue contribution by category of the cosmetics dataset

Step 7: Find Top 10 Brands for All Skin Types

We create a new calculated field:

[Combination]. [Dry]. [normal] + [Sensitive]

Then we drag the Brand to the Rows and the calculated field to the Columns.

We sort it in descending order. Apply a Top Filter to show the top 10 brands.

The result of this analysis displays the brands that offer products for multiple skin types.

Step 8: Find Top 10 Brands by Product Count

We drag the Brand to the Rows and the Product Name to the Columns.

We select COUNT(Product Name). Sort it in descending order.

Then we filter the 10 brands.

The result of this analysis identifies the brands with the number of products.

Step 9: Compare Average Prices Across Brands

We drag the Brand to the Rows and the Price to the Columns.

We change the aggregation to AVG(Price). Sort it in descending order.

Then we add labels to the visualization.

The result of this analysis compares the product prices across brands.

Step 10: Analyze Category- Distribution of Combination Skin Products

We drag the Category to the Rows and the Combination to the Columns.

We use SUM(Combination). Choose a Bar Chart.

The result of this analysis shows the availability of products for combination skin.

Step 11: Find Top Brands and Cosmetic Counts

We drag the Brand to the Rows and the Product Name to the Columns.

We use COUNT(Product Name). Sort it in descending order.

Then we display the N brands.

The result of this analysis highlights the leading brands based on product volume.

Step 12: Analyze Category Level Product Distribution

We drag the Category to the Rows and the Product Name to the Columns.

We use COUNT aggregation. Apply color based on count.

Then we add labels to the visualization.

The result of this analysis provides a comparison of product distribution among categories.

Step 13: Build the Dashboard

We click on New Dashboard and set the dashboard size.

Then we drag all the worksheets into the dashboard including:

- * Category Product Distribution
- * Category Revenue
- * Top 10 Brands (All Skin Types)
- * 10 Brands, by Product Count
- * Average Brand Price
- * Combination Skin Distribution
- * Top Brands and Cosmetic Counts

We arrange the visualizations logically.

Step 14: Add Interactive Filters

We click on the filter dropdown and select Apply to Worksheets.

We add filters for Category, Brand and Skin Type. Enable dashboard actions.

The result of this analysis is that all the charts update dynamically based on user selections of the cosmetics dataset.

Step 15: Validate Performance

We test each filter interaction. Verify the response time.

We check the dashboard rendering. Confirm that all the charts update correctly.

The performance summary is as follows:

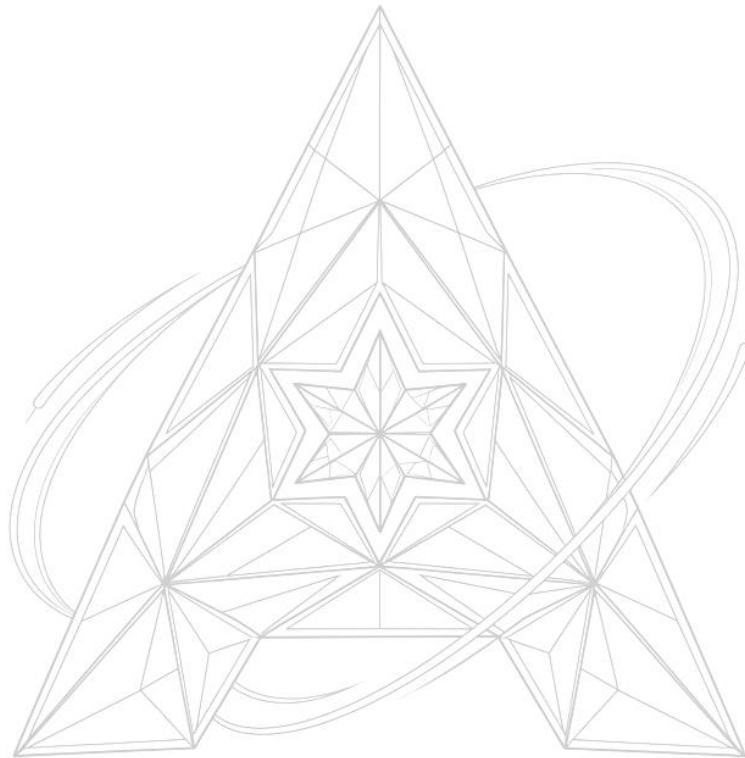
The dataset size is 1,472 rows and 10 columns.

The data volume is 1-2 MB.

Tableau handles filtering and aggregation efficiently.

The visualizations render smoothly within response times.

The interactive filters provide exploration without noticeable lag.



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— INNOVATORS —
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